

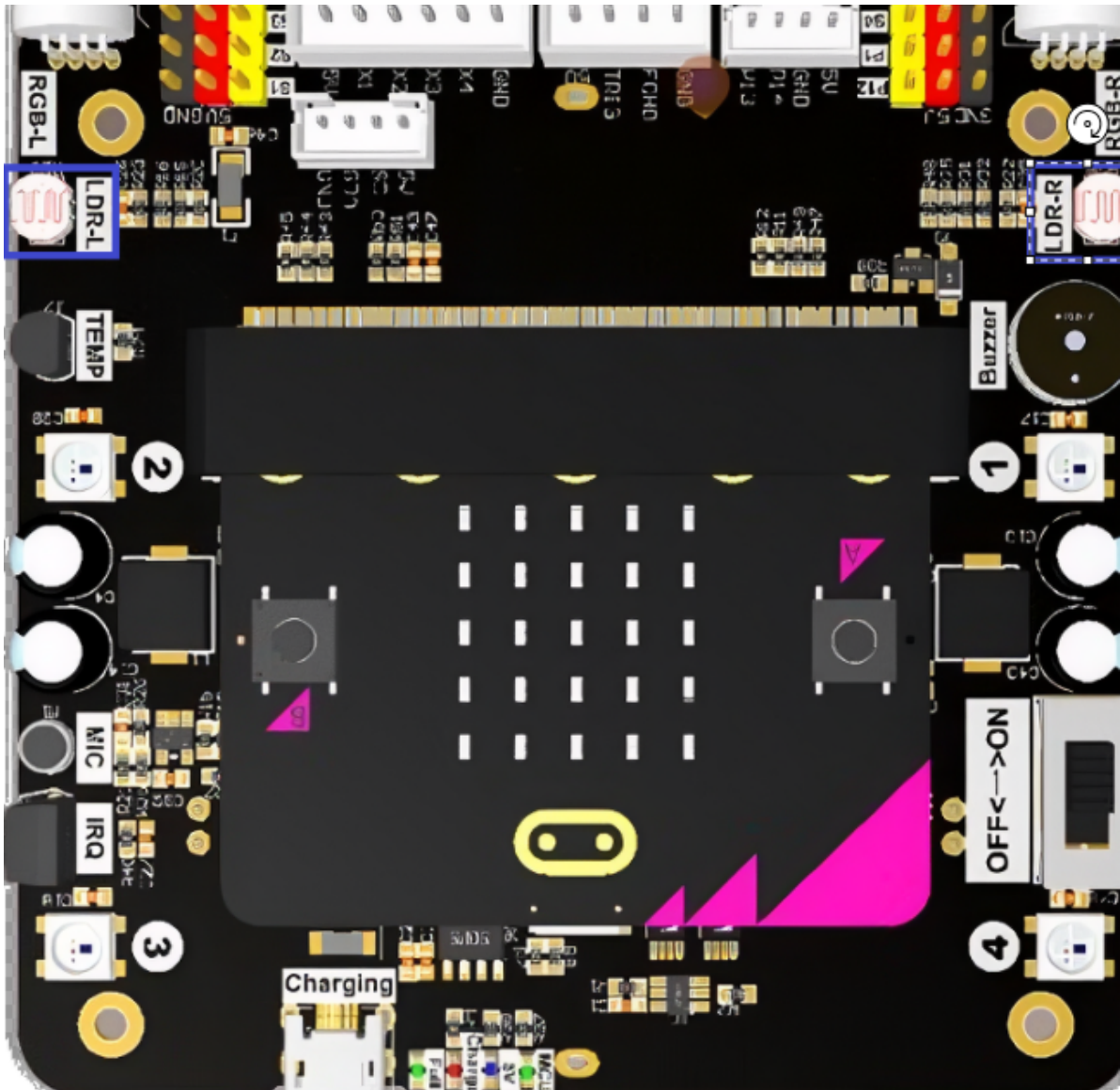
19. Moon Guardian

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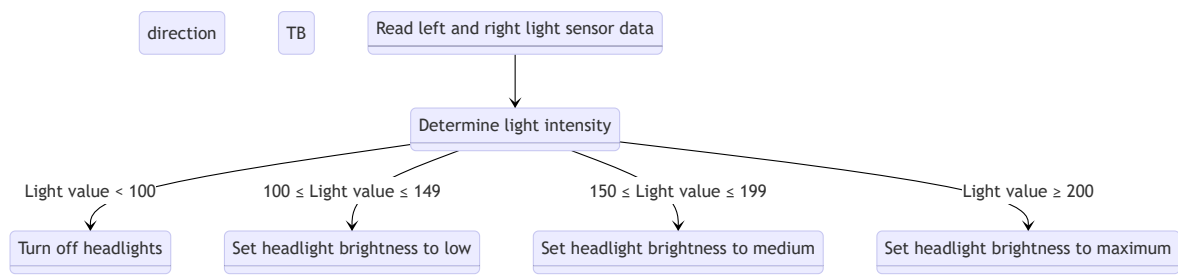
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Function Description

In this section, we use the light sensor to detect ambient brightness and implement the smart fill light function. When the ambient light is dim, the fill light is automatically turned on and its brightness is adjusted according to the light intensity; when the light is sufficient, the fill light is turned off.

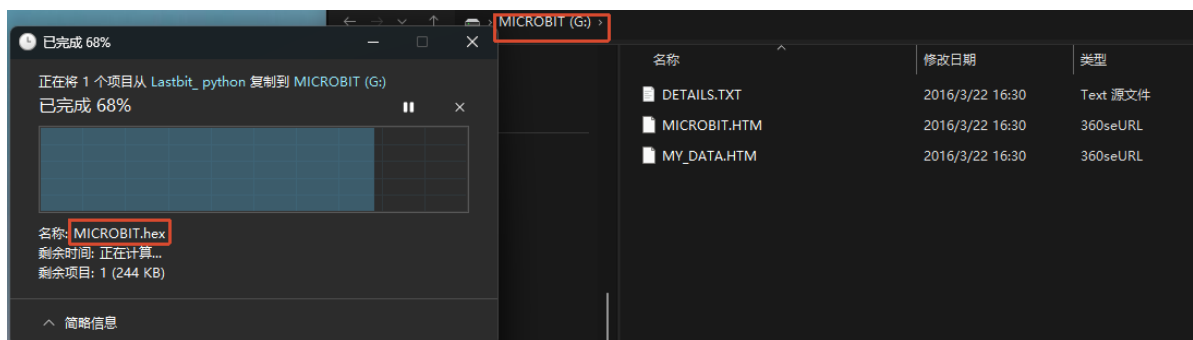


Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



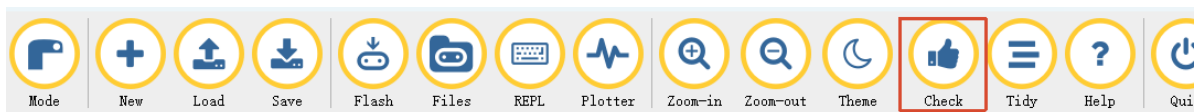
1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.
Click the `New` button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

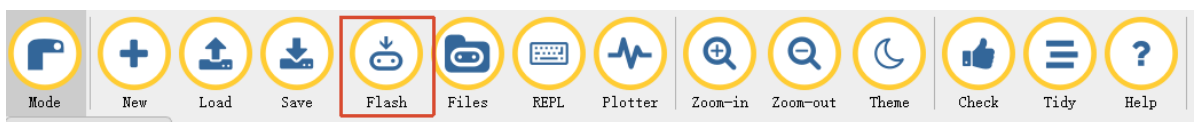
4. Now you can click `Check` to verify whether there are any problems with the code format. This case involves multi-line display patterns that may trigger warnings, but this does not affect flashing or running!



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows `Copied code onto micro:bit.` at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 19. Moon Guardian
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Left and right light colors
left_rgb = (0, 0, 0)
right_rgb = (0, 0, 0)
```

```

while True:
    # Read light sensors
    light_data = Lastbit.read_light()
    if light_data is not None:
        light_l = light_data[0]
        light_r = light_data[1]
        print("L:", light_l, "R:", light_r, "\n")
        average_light = (light_l + light_r) / 2

    # Threshold may need adjustment due to factory variations in light-
    # dependent resistors

    # Left light control
    if light_l >= 200:
        left_rgb = (255, 255, 255)
    elif light_l >= 150:
        left_rgb = (160, 160, 160)
    elif light_l >= 100:
        left_rgb = (60, 60, 60)
    else:
        left_rgb = (0, 0, 0)

    # Right light control
    if light_r >= 200:
        right_rgb = (255, 255, 255)
    elif light_r >= 150:
        right_rgb = (160, 160, 160)
    elif light_r >= 100:
        right_rgb = (60, 60, 60)
    else:
        right_rgb = (0, 0, 0)

    Lastbit.set_headlight_left_right(left_rgb, right_rgb)

    # Display brightness level
    if average_light > 40 and average_light < 70:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "00000:"
                           "99999"))
    elif average_light >= 70 and average_light < 110:
        display.show(Image("00000:"
                           "00000:"
                           "00000:"
                           "00900:"
                           "99999"))
    elif average_light >= 110 and average_light < 160:
        display.show(Image("00000:"
                           "00000:"
                           "00900:"
                           "09990:"
                           "99999"))
    elif average_light >= 160 and average_light < 170:
        display.show(Image("00000:"
                           "00900:"
                           "09990:"

```

```

        "99999:"
        "99999"))
    elif average_light >= 170 and average_light < 200:
        display.show(Image("00900:"
            "00900:"
            "99999:"
            "99999:"
            "99999"))
    elif average_light >= 200 and average_light < 240:
        display.show(Image("00900:"
            "09990:"
            "99999:"
            "99999:"
            "99999"))
    elif average_light >= 240:
        display.show(Image("99999:"
            "99999:"
            "99999:"
            "99999:"
            "99999"))
    else:
        display.clear()

sleep(100)

```

Experimental Phenomenon

Note here: make sure your hands are dry. Contact with water may cause a short circuit on the main board!

Make sure your hands are dry. Contact with water may cause a short circuit on the main board! Since the light sensors have errors, you can modify the thresholds of the light sensors in this case according to the actual situation. You can gradually cover the two light sensors with your hands, and you will notice the lights gradually getting brighter. You can also take the car to a darker environment to test. The darker the environment, the higher the pattern displayed on the Micro:bit matrix screen. Conversely, if you shine a flashlight on the light sensors, the car will turn off the lights, and the pattern displayed on the Micro:bit matrix screen will be lower.

Since the light sensors have errors, you can modify the thresholds of the light sensors in this case according to the actual situation.

